

MTH-395-00?: Research
Fall 2026

Classroom: Office, Howley 222, MWF 10-11AM

Professor: Michael Chou

Email: mchou@providence.edu

Course Description: In this research course, the students will be involved in designing an algorithm to achieve strong performance in multiplayer poker. We will analyze how reinforcement learning can be used to learn through self-play, study recent progress in the realm of multiplayer games, and measure the effectiveness of player exploiting strategies compared to Nash Equilibrium strategies.

Course Grading Standards: The course will be graded on a weekly goal basis. The student will be assessed on managing weekly goals. The students will give a grade justification at the end of the semester and arrive at a letter grade in collaboration with the professor.

- **Presentation:** Students will create an interactive exhibit to be presented during the April Celebration of Scholarship.
- **Project Paper:** Students will prepare a paper that summarizes the semester's work. This paper will include an "Introduction" section which contains a narrative version of the data audit a "Methodology" section that includes details of models used, a "Results" section that discusses model output and validation, and a "Conclusions" section that frames results in terms of the posed question.

Academic Integrity: Academic integrity is of the utmost importance in maintaining the high standards of scholarship in our community. Academic dishonesty is considered to be a serious offense against the community and represents a significant breach of trust between the professor, the classmates, and the student. There are many forms of academic dishonesty including plagiarism, falsifying data, misrepresenting class attendance, submitting the same work in two courses without prior approval, unauthorized discussion or distribution of exams or assignments, and offering or receiving unauthorized aid on exams or graded assignments. Violations of academic integrity will not be tolerated and will typically result in a 0 for the assignment up to an F for the course depending on the severity of the infraction.

Credit Hour Definition:

Academic engagement totaling at least 1 hour in class plus 2 hours out of class over the course of approximately 15 weeks [or the equivalents], for each 1 hour of credit should be included as an instructor expectation. Student will meet and work jointly with professor for 3 hours a week, and is expected to spend 6 hours outside of the class on the project.

Special Accommodations: If you have a documented disability and have been approved for academic accommodations, please contact me privately during my office hours as early as possible in the semester. All accommodations must be arranged through the Office of Academic Services. You may reach Disability Support at 865-1121.

Diversity and Inclusion: It is my intent to provide students from all backgrounds the best course possible. The diversity students bring into the class is a resource, a strength,

and a benefit. I will make every effort to present material and activities that are respectful of your various identities, including (but not limited to) those tied to ethnicity, race, gender, religion, sexuality, disability, age, socioeconomic status, and culture. It is critical that we work together to create a safe and bias-free learning environment in which all students can thrive and succeed. I take this mission very seriously and your suggestions on how to best accomplish it are always encouraged and appreciated.

Mental Health: As a student, you may experience a range of challenges that can interfere with learning, such as strained relationships, increased anxiety, substance use, feeling down, difficulty concentrating and/or lack of motivation. These mental health concerns or stressful events may affect your ability to attend class, concentrate, complete work, take an exam, or participate in daily activities. If mental health and adjustment concerns are causing distress, please speak to me and/or reach out for personal support. Remember that asking for help is a sign of strength and courage.